

Guided Assignment 2

Hints and worked solutions

Use sparingly. Try the student assignment first. This opening page contains hints only. Worked answers begin on the next page. After checking a solution, close this file and reconstruct the reasoning yourself. Correct reasoning and labelled qualitative sketches matter more than matching wording.

One hint per stage

1. Add and subtract the two activity expressions. In the general normal equation, the matrix multiplying D must be $n \times n$.
2. Threshold requires $e^{-t/\tau_{RC}} = 1 - 1/J$. First find the currents needed for the two target rates, then fit the affine current equation.
3. The negative-frequency partner must cancel the imaginary part of the reconstructed signal. RMS rescales linearly when a trace is multiplied by a positive constant.
4. Each spike starts a fresh copy of the same kernel. The second spike adds to the still-decaying response from the first. Track units before inserting a time-step factor.
5. Substitute $R = X + \eta$ into both averaged products; terms correlating signal with noise vanish. A negative kernel lag samples a response later than the current estimate time.
6. Changing the decoded function changes the target matrix, not the already-measured activity matrix. Any linear combination of Stage 1's activities is affine in x .
7. Expand the decoded vector coordinate by coordinate inside the receiving neuron's dot product. For multiplication, compare how changing x affects the result at two different values of y .
8. No new technique is required. If one part stalls, identify which earlier stage supplied the missing idea and redo a nearby exercise before returning.

1. Encoding and decoding foundations

1a. The table is

x	J_1	J_2	a_1	a_2
-1	0	2	0	2
0	1	1	1	1
1	2	0	2	0

The curves are straight lines $a_1 = x + 1$ and $a_2 = 1 - x$ throughout this domain. Equal positive activity can cancel after applying opposite-signed decoder coefficients. Activity is not the represented value.

1b. At $x = 0$, $d_1 + d_2 = 0$; at $x = 1$, $2d_1 = 1$. Thus $D = (1/2, -1/2)$ and $\hat{x} = (a_1 - a_2)/2 = x$. At $x = 1/2$, the activities are $(3/2, 1/2)$ and $\hat{x} = 1/2$. A negative coefficient subtracts a neuron's contribution; the firing rate remains nonnegative.

1c. The shapes are $A : n \times N$, $X : d \times N$, $D : d \times n$. Differentiating the objective gives

$$D(AA^T + \lambda I_n) = XA^T, \quad D = XA^T(AA^T + N\sigma^2 I_n)^{-1}.$$

For independent activity noise with variance σ^2 , its contribution to scalar output variance is $\sigma^2 \sum_i d_i^2$. Shrinking coefficients reduces this sensitivity, but constrains how well the decoder can fit clean targets. Regularization supplies a tradeoff, not a guarantee of improvement on every noise realization.

2. LIF dynamics

2a. For $J = 0.5$, voltage rises concavely toward 0.5 and never spikes. For $J = 1$, it approaches threshold asymptotically and never reaches it in finite time in the continuous model. For $J = 2$, it crosses threshold in finite time, spikes, resets, stays at zero for 2 ms, and repeats its charging curve. Show spikes separately from voltage; voltage is not a Dirac impulse.

2b. Solving $1 = J(1 - e^{-t_{charge}/\tau_{RC}})$ yields

$$t_{charge} = -\tau_{RC} \ln(1 - 1/J), \quad T_{ISI} = \tau_{ref} - \tau_{RC} \ln(1 - 1/J), \quad G(J) = T_{ISI}^{-1}.$$

The first spike from a non-refractory reset state waits only t_{charge} ; later intervals include the preceding refractory hold. For $J = 2$, $t_{charge} = 20 \ln 2 \approx 13.9$ ms and the steady rate is about 63.0 Hz. For $J \leq 1$, the rate is zero.

2c. Use seconds and rates in Hz:

$$J(a) = \frac{1}{1 - \exp[(\tau_{ref} - 1/a)/\tau_{RC}]}.$$

Hence $J(40) \approx 1.463$, $J(150) \approx 4.805$, giving $J^{bias} = J(40)$ and $\alpha = J(150) - J(40) \approx 3.342$. At $x = 0$, both partners receive J^{bias} and have steady rates of 40 Hz. At $x = 1$, the positive partner fires near 150 Hz; the negative partner receives $J \approx -1.879$ and is silent under the zero-voltage clamp. The sine input is 5 Hz: positive-half cycles favour the positive neuron and negative-half cycles favour its partner; at zero crossings both have the same instantaneous rate-model prediction. Actual spike timing depends on voltage history, so do not interpret the rate curve as exact instantaneous spike times.

2d. Outside refractory periods,

$$v_{k+1} = v_k + \frac{\Delta t}{\tau_{RC}}(J_k - v_k).$$

Here $0.4 + (1/20)(2 - 0.4) = 0.48$. Euler approximation and threshold/refractory timing rounded to the grid can alter the simulated period; a smaller step reduces these errors. Finite observation windows, initial voltage, and incomplete last cycles also affect counts. A smaller step cannot remove the fact that a finite window contains an integer count with boundary and startup effects.

3. Random signals

3a. There are $N = T/\Delta t = 1000$ samples; $\Delta f = 1/T = 1$ Hz and $\Delta\omega = 2\pi/T = 2\pi$ rad/s. Reality requires $X(-3 \text{ Hz}) = X(+3 \text{ Hz})^* = 2 - 3i$. Set the DC coefficient to zero and coefficients at $|f| > 8$ Hz to zero. The Nyquist coefficient is also zero here since it lies outside the band.

3b. The average magnitude is roughly flat across the retained nonzero frequency bins, vanishes at DC by construction, and vanishes beyond $\omega = \pm 16\pi$ rad/s. Individual magnitudes fluctuate. Multiply the trace by $0.5/0.8 = 5/8$. RMS is not a peak bound: a sine with amplitude $1/\sqrt{2}$ has RMS $1/2$ but peaks at about ± 0.707 .

3c. The higher cutoff permits faster changes and shorter temporal correlations. Both traces retain the same mean-square amplitude 0.25 after normalization; individual peaks need not match. A fixed low-pass filter attenuates higher frequencies more, so equal RMS does not imply equal tracking distortion. For a flat power spectrum, spreading the same total power over a wider band reduces its power per frequency bin on average.

4. PSC filtering

4a. The normalizers are

$$c_0 = \int_0^\infty e^{-t/\tau} dt = \tau, \quad c_1 = [-\tau t e^{-t/\tau}]_0^\infty + \tau \int_0^\infty e^{-t/\tau} dt = \tau^2.$$

Optionally $c_2 = 2\tau^3$; generally $c_n = n!\tau^{n+1}$. The $n = 0$ kernel starts at $1/\tau$ and peaks at zero. The $n = 1$ kernel starts at zero, rises, peaks at $t = \tau$ with height $1/(e\tau)$, then decays. Both vanish at negative times. Unit area gives DC gain one: a constant input present for long enough passes unchanged after startup. It does not preserve arbitrary time-varying RMS or energy.

4b. The filtered activity is $s(t) = h_0(t) + h_0(t - \tau)$. Since $\tau = 0.010$ s, $1/\tau = 100 \text{ s}^{-1}$. The values are 100 Hz at 0^+ , $100e^{-1} \approx 36.8$ Hz at τ^- , and $100(1 + e^{-1}) \approx 136.8$ Hz at τ^+ . Sketch two upward jumps, each followed by exponential decay. Superposition includes the tail of every previous spike.

4c. Substitution gives

$$s_{ik} \approx \sum_{\ell=0}^k b_{i\ell} h_{k-\ell}.$$

There is no extra time-step factor with counts and samples of a continuous kernel. With impulse heights, the Riemann-sum factor Δt is needed and cancels their $1/\Delta t$. The output s has rate units, so a rate decoder acts directly on it. Another kernel convention, such as discrete coefficients already multiplied by Δt , requires matching normalization. Never infer a universal output multiplier without checking how spikes and kernels were stored.

4d. Increasing τ broadens the response, typically reduces fast spike ripple, and increases smoothing, lag, and attenuation of rapid signal components. Increasing n at fixed τ also suppresses more high-frequency content and delays the response: the peak occurs at $n\tau$ for $n > 0$, and the kernel's mean delay is $(n+1)\tau$. The $n = 1$ onset rises continuously rather than jumping. Slow inputs usually suffer less filtering distortion, but can retain spike noise and static decoder error. These trends do not prove total RMSE improves in every case.

4e. At 2 Hz, $\omega\tau \approx 0.2512$ and $|H| \approx 0.970$. At 10 Hz, $\omega\tau \approx 1.256$ and $|H| \approx 0.623$. The faster component is much more attenuated; the filter also shifts its phase. These gains describe the linear filter alone, not a guarantee about the whole neural network.

5. Optimal filtering

5a. The simplified model gives

$$\langle XR^* \rangle = S_X, \quad \langle |R|^2 \rangle = S_X + S_\eta, \quad H = \frac{S_X}{S_X + S_\eta}.$$

The numerator measures response content correlated with the input, while the denominator includes signal and noise. Large response magnitude can mean large noise. This simplified expression assumes unit response gain; the general cross-spectrum formula also accounts for gain and phase relationships.

5b. The gains are 0.9 and 0.1. For zero signal power and positive noise power, the gain is zero. If both powers are zero, the formula is 0/0: the objective gives no constraint at that frequency because the response vanishes. Setting $H = 0$ is harmless there.

5c. The input has support only within $\pm 12\pi$ rad/s, with DC absent for a zero-mean construction. The response also contains broadband spiking content. Under the stated ideal ensemble assumptions, the decoded spectrum has no out-of-band support where response noise is uncorrelated with the input. It can still have in-band noise and attenuation. Finite-data estimates need not give exact zeros; individual random spectra are not smooth plateaus.

5d. At lag $u = -5$ ms and estimate time $t = 100$ ms, the convolution uses $r(105$ ms). That is future information. Offline analysis has it, but a live synapse does not. A causal PSC with zero response at negative lag is an alternative; its smoothing and delay can increase distortion relative to the unconstrained optimum. An optimal filter need not always be acausal, but unconstrained optimality does not ensure causality.

6. Function targets and transfer

6a. The target row is $F = (-1, 1, 3)$. Matching endpoints gives coefficients $(p, q) = (3/2, -1/2)$; the midpoint also matches. At $x = -1/2$, activities are $(1/2, 3/2)$ and the decoded value is zero, as desired. For a general q -dimensional target,

$$D^f = FA^T(AA^T + N\sigma^2 I_n)^{-1} \in \mathbb{R}^{q \times n}.$$

The exact toy answer is unregularized; positive regularization generally changes it.

6b. For targets $(1, 0, 1)$, endpoint matching requires $p = q = 1/2$, but the midpoint then decodes to 1 rather than 0. Every combination here is

$$p(x+1) + q(1-x) = (p-q)x + (p+q),$$

an affine function, so it cannot equal x^2 on an interval. A decoder is linear in activities, not necessarily in the represented input: nonlinear tuning curves can provide nonlinear basis functions. Additional suitable tuning curves can improve approximation, though a finite arbitrary population does not guarantee exact computation.

6c. Use $s_i(t) = \sum_k h(t - t_i^k)$ and $\hat{x}(t) = \sum_i d_i s_i(t)$, or $\hat{\mathbf{x}} = D\mathbf{s}$. Useful transfer needs matching neuron tuning/current parameters, consistent units and filter normalization, inputs within the trained domain, and bandwidth compatible with the filter and neuronal dynamics. Filtered spikes fluctuate around the rate approximation; filtering and voltage dynamics add temporal distortion. Exact instant-by-instant equivalence is not promised.

6d. (i) Static decoder distortion or regularization bias: improve basis coverage or fitting and review regularization. (ii) Spike variability: more suitable neurons and/or more smoothing can help. (iii) Causal filtering and neural dynamics: reducing time constants may sharpen the response at the cost of ripple. (iv) Domain mismatch: enlarge or rescale the representation and train accordingly. More neurons can improve approximation and averaging but do not automatically remove filtering lag or repair the trained range. These are likely diagnoses under correct units and implementation, not unique explanations of every observed trace.

7. Connections and architectures

7a. Expanding gives

$$J_i = \sum_{j=1}^n \underbrace{\left(\alpha_i \sum_{k=1}^q e_{ik} D_{kj}^f \right)}_{w_{ij}} a_j + J_i^{bias}.$$

Define $E_{ik} = \alpha_i e_{ik}$, so $E \in \mathbb{R}^{m \times q}$ and $W = ED^f \in \mathbb{R}^{m \times n}$. Each receiving neuron's bias is added once outside the activity-weighted sum. A function decoder's constant target term is distinct from this neuronal bias. With a common synaptic filter, replace rate activities by filtered spike activities in the weighted current expression.

7b. The gain-scaled encoders are $E = (2, -3)^T$, hence

$$W = \begin{bmatrix} 3 & -1 \\ -9/2 & 3/2 \end{bmatrix}.$$

Decoding $(1, 1)^T$ gives $\hat{y} = 1$. Encoding it gives currents $(2 \cdot 1 + 1, -3 \cdot 1 + 2)^T = (3, -1)^T$. Multiplying by W and adding bias gives the same result. Positive weights are 3 and 3/2; negative weights are -1 and $-9/2$. A negative current is allowed in the current equation; the LIF voltage clamp and nonlinearity govern the subsequent neural response.

7c. The ramp has $x \in [-1, 0]$, so $y \in [-1, 1]$. The sinusoid gives $y \in [0.6, 1.4]$, exceeding radius 1. A radius of at least 1.4 covers the desired values, with practical margin useful when accommodating approximation error. A larger radius requires corresponding tuning parameters and decoder fitting. The expected failure outside the domain is reduced accuracy, not necessarily literal clipping. A filtered step rises gradually and may have superimposed spike jitter. Fit over the intended domain so the connection represents the function on new inputs rather than only a particular waveform.

7d. Use scalar populations X, Y, Z , with $X \xrightarrow{x \mapsto 0.5x} Z$ and $Y \xrightarrow{y \mapsto 2y} Z$. Each represents one scalar. For scalar receiving encoder e_i ,

$$J_i = \sum_j \alpha_i e_i D_j^{0.5x} a_j^X + \sum_k \alpha_i e_i D_k^{2y} a_k^Y + J_i^{bias}.$$

Add the receiving bias once, regardless of the number of incoming populations. In spiking mode each branch's activities must be appropriately filtered.

7e. Route X into coordinate 1 of a joint population U through $x \mapsto (x, 0)^T$, and Y into coordinate 2 through $y \mapsto (0, y)^T$. Population U represents a two-dimensional vector. Fit an outgoing function decoder with targets $u_1 u_2$ over its intended joint domain, then feed a scalar output population P . The input and output populations are one-dimensional. Choose U 's domain to cover the intended pairs; a unit-radius disk does not cover every corner of $[-1, 1]^2$.

If $xy = g(x) + k(y)$, setting $y = 0$ forces g to be constant. Setting $y = 1$ then requires that same g to vary as x , a contradiction on a domain including these values. More generally, differences between two distinct

y values would have to be both independent of x and proportional to x . If only $x + y$ is represented, pairs $(1, 0)$ and $(1/2, 1/2)$ both give sum 1 but products 0 and $1/4$. No function of that sum alone can recover both products. Joint nonlinear tuning preserves the information and supplies interaction basis functions. This architecture supports approximation rather than guaranteeing an exact finite-population product.

8. Readiness check: compact key

8a. Cutoffs are $\omega = \pm 24\pi$ rad/s, with roughly flat average magnitude in the retained bins and zero DC. At 3 Hz the trace varies more slowly with unchanged RMS; a fixed low-pass PSC generally introduces less signal distortion, though spike noise remains.

8b. At $x = -1/2$, $J_+ = 0.5$ and $J_- = 1.5$. The positive-encoder voltage rises toward 0.5 and is silent. The negative-encoder neuron repeatedly reaches threshold, resets, and holds at zero for 2 ms. Its first charging time is $20 \ln 3$ ms; later intervals add the refractory hold. Exact timing is not required.

8c. The normalizer is τ^2 , so $h(t) = te^{-t/\tau}/\tau^2$ for $t \geq 0$. Its derivative vanishes at $t = \tau$, the maximum. Larger τ reduces fast ripple but increases delay and distortion of rapidly changing signals.

8d. The decoder is

$$D = XA^T(AA^T + 200\sigma^2 I_{30})^{-1} \in \mathbb{R}^{2 \times 30}, \quad s_{ik} \approx \sum_{\ell=0}^k b_{i\ell} h_{k-\ell}.$$

There is no extra Δt in this count-based convention.

8e. The scaled encoder matrix is $E : 50 \times 2$, so $W = ED^f : 50 \times 30$ and $w_{ij} = \alpha_i \sum_{k=1}^2 e_{ik} D_{kj}^f$. For subtraction, route scalar X to scalar Z with identity and scalar Y to Z with multiplication by -2 . For multiplication, route the two scalar inputs into the two coordinates of a joint two-dimensional population, then decode the product into a scalar output population.

8f. The desired range is $[-1/2, 5/2]$, requiring a symmetric radius of at least $5/2$ if that convention is used. A domain error can persist under a constant out-of-range input. Lag describes a delayed or smoothed response to changes and can occur even when all targets stay within range.

What to revisit

Dimensional errors: Stage 1 or 7. Voltage/spike confusion: Stage 2. Hz versus rad/s or RMS/radius confusion: Stage 3. Time-step scaling or smoothing tradeoff: Stage 4. Cross-spectra or causality: Stage 5. Changing training targets or interpreting error: Stage 6. Addition versus multiplication: Stage 7. Redo the corresponding readiness item before attempting the existing practice test.